As an example, we can assume a Gaussian guider with fixed variance $\sigma^2 = 1$, such that

$$\log p_\Phi(y|z_t) = -\frac{1}{2}\|y - T(\mathcal{D}(z_0(z_t)))\|_2^2 \tag{17}$$

becomes a L_2 regression objective.

Fig. 14 demonstrates how this formulation can serve as an upsampling mechanism of an unconditional model trained on 256^2 images, where unconditional samples of size 256^2 guide the convolutional synthesis of 512^2 images and T is a $2\times$ bicubic downsampling. Following this motivation, we also experiment with a perceptual similarity guiding and replace the L_2 objective with the LPIPS [106] metric, see Sec. 4.4.

D. Additional Results

D.1. Choosing the Signal-to-Noise Ratio for High-Resolution Synthesis

Figure 15. Illustrating the effect of latent space rescaling on convolutional sampling, here for semantic image synthesis on landscapes. See Sec. 4.3.2 and Sec. D.1.

As discussed in Sec. 4.3.2, the signal-to-noise ratio induced by the variance of the latent space (*i.e.* $\mathrm{Var}(z)/\sigma_t^2$) significantly affects the results for convolutional sampling. For example, when training a LDM directly in the latent space of a KL-regularized model (see Tab. 8), this ratio is very high, such that the model allocates a lot of semantic detail early on in the reverse denoising process. In contrast, when rescaling the latent space by the component-wise standard deviation of the latents as described in Sec. G, the SNR is descreased. We illustrate the effect on convolutional sampling for semantic image synthesis in Fig. 15. Note that the VQ-regularized space has a variance close to 1, such that it does not have to be rescaled.

D.2. Full List of all First Stage Models

We provide a complete list of various autoenconding models trained on the OpenImages dataset in Tab. 8.

D.3. Layout-to-Image Synthesis

Here we provide the quantitative evaluation and additional samples for our layout-to-image models from Sec. 4.3.1. We train a model on the COCO [4] and one on the OpenImages [49] dataset, which we subsequently additionally finetune on COCO. Tab 9 shows the result. Our COCO model reaches the performance of recent state-of-the art models in layout-to-image synthesis, when following their training and evaluation protocol [89]. When finetuning from the OpenImages model, we surpass these works. Our OpenImages model surpasses the results of Jahn et al [37] by a margin of nearly 11 in terms of FID. In Fig. 16 we show additional samples of the model finetuned on COCO.

D.4. Class-Conditional Image Synthesis on ImageNet

Tab. 10 contains the results for our class-conditional LDM measured in FID and Inception score (IS). LDM-8 requires significantly fewer parameters and compute requirements (see Tab. 18) to achieve very competitive performance. Similar to previous work, we can further boost the performance by training a classifier on each noise scale and guiding with it,

f	$\lvert\mathcal{Z}\rvert$	c	**R-FID** ↓	**R-IS** ↑	**PSNR** ↑	**PSIM** ↓	**SSIM** ↑
16 *VQGAN* [23]	16384	256	4.98	–	19.9 ± 3.4	1.83 ± 0.42	0.51 ± 0.18
16 *VQGAN* [23]	1024	256	7.94	–	19.4 ± 3.3	1.98 ± 0.43	0.50 ± 0.18
8 *DALL-E* [66]	8192	-	32.01	–	22.8 ± 2.1	1.95 ± 0.51	0.73 ± 0.13
32	16384	16	31.83	40.40 ± 1.07	17.45 ± 2.90	2.58 ± 0.48	0.41 ± 0.18
16	16384	8	5.15	144.55 ± 3.74	20.83 ± 3.61	1.73 ± 0.43	0.54 ± 0.18
8	16384	4	1.14	201.92 ± 3.97	23.07 ± 3.99	1.17 ± 0.36	0.65 ± 0.16
8	256	4	1.49	194.20 ± 3.87	22.35 ± 3.81	1.26 ± 0.37	0.62 ± 0.16
4	8192	3	0.58	224.78 ± 5.35	27.43 ± 4.26	0.53 ± 0.21	0.82 ± 0.10
4†	8192	3	1.06	221.94 ± 4.58	25.21 ± 4.17	0.72 ± 0.26	0.76 ± 0.12
4	256	3	0.47	223.81 ± 4.58	26.43 ± 4.22	0.62 ± 0.24	0.80 ± 0.11
2	2048	2	0.16	232.75 ± 5.09	30.85 ± 4.12	0.27 ± 0.12	0.91 ± 0.05
2	64	2	0.40	226.62 ± 4.83	29.13 ± 3.46	0.38 ± 0.13	0.90 ± 0.05
32	KL	64	2.04	189.53 ± 3.68	22.27 ± 3.93	1.41 ± 0.40	0.61 ± 0.17
32	KL	16	7.3	132.75 ± 2.71	20.38 ± 3.56	1.88 ± 0.45	0.53 ± 0.18
16	KL	16	0.87	210.31 ± 3.97	24.08 ± 4.22	1.07 ± 0.36	0.68 ± 0.15
16	KL	8	2.63	178.68 ± 4.08	21.94 ± 3.92	1.49 ± 0.42	0.59 ± 0.17
8	KL	4	0.90	209.90 ± 4.92	24.19 ± 4.19	1.02 ± 0.35	0.69 ± 0.15
4	KL	3	0.27	227.57 ± 4.89	27.53 ± 4.54	0.55 ± 0.24	0.82 ± 0.11
2	KL	2	0.086	232.66 ± 5.16	32.47 ± 4.19	0.20 ± 0.09	0.93 ± 0.04

Table 8. Complete autoencoder zoo trained on OpenImages, evaluated on ImageNet-Val. † denotes an attention-free autoencoder.

layout-to-image synthesis on the COCO dataset

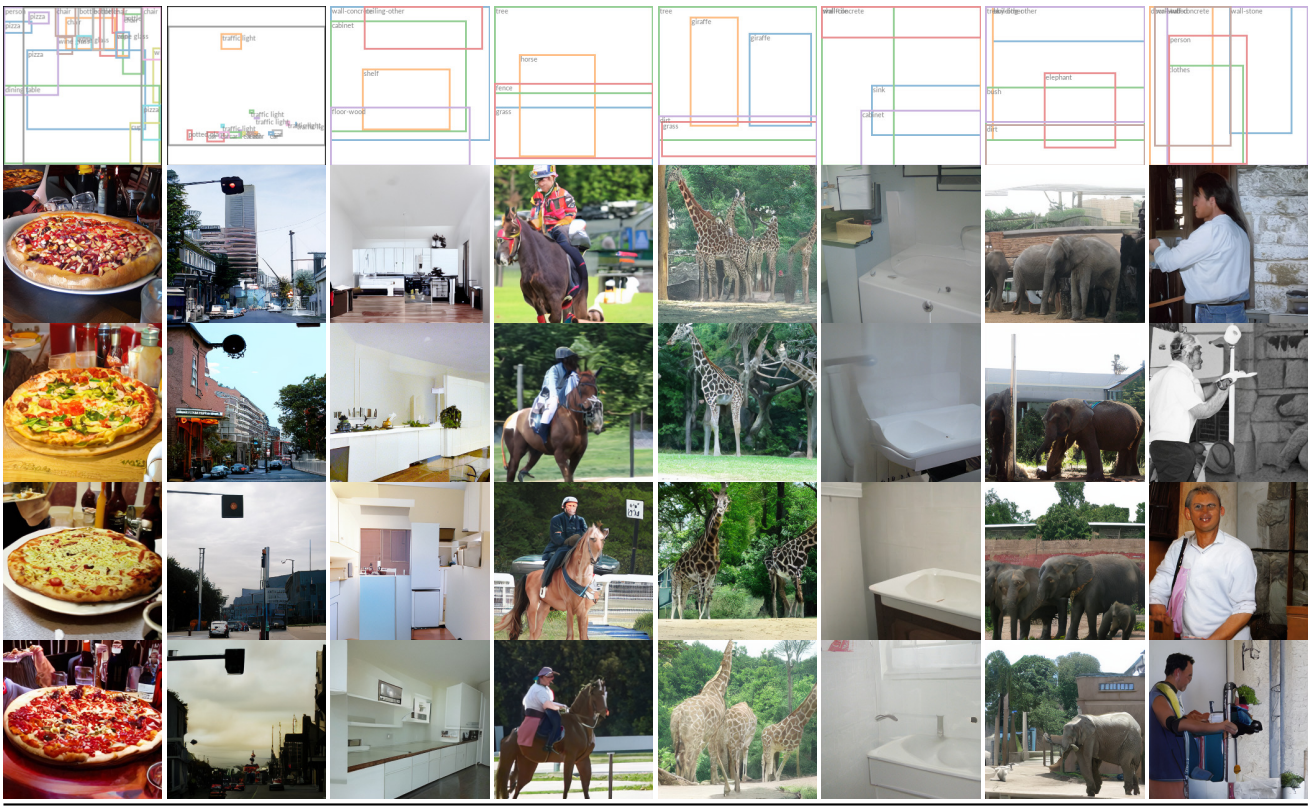

Figure 16. More samples from our best model for layout-to-image synthesis, *LDM-4*, which was trained on the OpenImages dataset and finetuned on the COCO dataset. Samples generated with 100 DDIM steps and $\eta = 0$. Layouts are from the COCO validation set.

see Sec. C. Unlike the pixel-based methods, this classifier is trained very cheaply in latent space. For additional qualitative results, see Fig. 26 and Fig. 27.